MULTI-CLOUD FDE STACK #03

MULTI-CLOUD FDE STACK · ISSUE 03 OF 08

VPC, firewall rules, VNet.

Most first deployments fail on the network, not the code.

03

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Akhilesh Ranjan Singh

Senior DevSecOps Engineer · github.com/iarsingh

Swipe →

Same idea, different rules

	SCOPE	FILTERING
AWS VPC	Regional	Security groups + NACLs
GCP VPC	Global	Firewall rules by tag or SA
Azure VNet	Regional	NSGs on subnet or NIC

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Akhilesh Ranjan Singh
github.com/iarsingh

Keep managed services private

01 AWS

VPC endpoints: gateway for S3, interface via PrivateLink.

02 GCP

Private Google Access or Private Service Connect.

03 Azure

Private endpoints with private DNS zones.

04 Egress

NAT gateway or Cloud NAT, with an allowlist.

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Akhilesh Ranjan Singh
github.com/iarsingh

Debugging a timeout

1. DNS resolves?
2. Route to target?
3. Egress rule allows 443?
4. Proxy or firewall?
5. curl -v from the pod

> Test from inside the workload. A laptop on the VPN proves nothing about the pod.

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Akhilesh Ranjan Singh
github.com/iarsingh

Network habits

DO

- ✓ Deny inbound by default
- ✓ Private endpoints for data
- ✓ Flow logs on during go-live
- ✓ Write down every allowlisted domain

AVOID

- ✗ Public database endpoints
- ✗ 0.0.0.0/0 to fix it fast
- ✗ Assume the LLM API is reachable
- ✗ Debug from your laptop

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github.com/iarsingh

Ask on day one

- [x] Is egress forced through a proxy?
- [x] Which domains are allowlisted?
- [x] Private endpoints required?
- [x] Who changes firewall rules?
- [x] How long does a change take?

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github.com/iarsingh

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KEY TAKEAWAY

Ask about egress before you write the first line of code.

→ Know regional vs global scope

→ Private endpoints for managed data

→ Debug from inside the pod

YOUR TURN

What is the longest you have waited for a firewall change?

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Next in the series: #04 Secrets: one pattern, three vaults

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Akhilesh Ranjan Singh
github.com/iarsingh